



JAVA BACKEND ENGINEER ROADMAP

- ✓ Step-by-Step Roadmap to Become an Interview-Ready Java Backend Developer
- ✓ All courses available on **TCS Udemy Business**



Start Your Journey
Goal: Backend Developer



Core Foundation
Strengthen Java & Programming

LEGEND

- Learning Phase
- Hands-on / Practice
- Project
- Certification
- Interview Prep

PHASE 1



Core Java
(2–3 Weeks)

PHASE 2



Spring Boot
(4 Weeks)

PHASE 3



Database & SQL
(1 Week)

PHASE 4



Git & GitHub
(3 Days)

PHASE 5



Microservices
(4–5 Weeks)

PHASE 6



AWS Cloud
(2 Weeks)

PHASE 7



Docker & Kubernetes
(1–2 Weeks)

PHASE 8



Testing
(1 Week)

PHASE 9



CI/CD
(1 Week)

PHASE 10



System Design
(3–4 Weeks)

COURSE

Java Programming Masterclass

Instructor: Tim Buchalka

🔗 Link:
tcsglobal.udemy.com/course/java-the-complete-java-developer-course/

COURSE

Spring Boot 4, Spring 7 & Hibernate for Beginners

Instructor: Chad Darby

🔗 Link:
tcsglobal.udemy.com/course/spring-hibernate-tutorial/

COURSE

SQL - Complete Database Course

(Choose any highly rated SQL course)

🔗 Browse:
tcsglobal.udemy.com/

COURSE

Git Complete: The Definitive Guide

Instructor: Jason Taylor

🔗 Link:
tcsglobal.udemy.com/course/git-complete/

COURSE

Master Microservices with Spring Boot & Kubernetes

Instructor: (Madan Reddy)

🔗 Link:
tcsglobal.udemy.com/course/master-microservices-with-spring-docker-kubernetes/

COURSE

Ultimate AWS Certified Cloud Practitioner

Instructor: Stephane Maarek

🔗 Link:
tcsglobal.udemy.com/course/aws-certified-cloud-practitioner-new/

COURSE

Docker & Kubernetes: The Practical Guide

Instructor: Maximilian Schwarzmüller

🔗 Link:
tcsglobal.udemy.com/course/docker-kubernetes-the-practical-guide/

COURSE

JUnit 5 & Mockito Masterclass

(Choose any highly rated course)

🔗 Browse:
tcsglobal.udemy.com/

COURSE

CI/CD with Jenkins & GitHub Actions

(Choose any highly rated course)

🔗 Browse:
tcsglobal.udemy.com/

COURSE / RESOURCES

System Design

(YouTube / Books / High Rated Course)

🔗 Browse:
tcsglobal.udemy.com/

YOU'LL LEARN

- Core Java
- OOPs
- Collections
- Multithreading
- Exception Handling
- Streams, Lambda
- JDBC

YOU'LL LEARN

- Spring Core
- Spring Boot
- Spring MVC
- Spring Data JPA
- Hibernate
- REST APIs
- Spring Security

YOU'LL LEARN

- Joins
- Indexes
- Transactions
- Views
- Stored Procedures
- Query Optimization

YOU'LL LEARN

- Git Basics
- Branching
- Merge vs Rebase
- Stash
- Pull Requests
- GitHub Workflows

YOU'LL LEARN

- Spring Cloud
- Eureka, Config Server
- API Gateway, OpenFeign
- Kafka, RabbitMQ
- Resilience4j
- Docker, Kubernetes
- Prometheus, Grafana
- OAuth2, Security

YOU'LL LEARN

- IAM
- EC2
- S3
- RDS
- VPC
- Lambda
- CloudWatch
- Route53

YOU'LL LEARN

- Docker Basics
- Docker Compose
- Kubernetes Basics
- Pods, Services
- Deployments
- Ingress, Helm

YOU'LL LEARN

- Unit Testing
- JUnit 5
- Mockito
- Integration Testing
- REST Assured

YOU'LL LEARN

- Jenkins Basics
- Pipelines
- GitHub Actions
- Build, Test
- Docker Integration
- Deployments

YOU'LL LEARN

- System Design Basics
- Scalability
- Load Balancing
- Caching (Redis)
- Database Sharding
- CAP Theorem
- Design Patterns

PROJECT

Library Management System

PROJECT

Employee Management System

PROJECT

E-Commerce Microservices Application

User, Product, Order, Inventory, Payment Services
Tech: Spring Boot, Spring Cloud, Kafka, Docker, Kubernetes, Redis

PROJECT

Deploy on AWS

EC2, S3, RDS, IAM, Docker, GitHub Actions
CI/CD Pipeline

PROJECT

Design a Scalable System

★ TECH STACK YOU WILL MASTER

- Java
- Spring Boot
- Spring Cloud
- Hibernate
- MySQL
- PostgreSQL
- Redis
- Kafka
- Docker
- Kubernetes
- AWS
- Git & GitHub
- JUnit
- Mockito
- CI/CD
- Linux

📅 WEEKLY PLAN (15–20 HRS)

- Mon** Course + Coding (2 hrs)
- Tue** Build Project (2 hrs)
- Wed** DSA + SQL (2 hrs)
- Thu** Course (2 hrs)
- Fri** Project (2 hrs)
- Sat** Project + Revision (5 hrs)
- Sun** LeetCode + System Design (4 hrs)

🎯 INTERVIEW PREPARE

- ✓ DSA & Coding Practice (LeetCode)
- ✓ SQL & Database Concepts
- ✓ Spring Boot & Microservices
- ✓ System Design
- ✓ Mock Interviews & Resume

📖 USEFUL RESOURCES

LeetCode – leetcode.com
HackerRank – hackerrank.com
System Design Primer – GitHub
Books: DDIA, System Design Interview
AWS Docs – docs.aws.amazon.com

🏆 SUCCESS FORMULA

CONSISTENCY
+
PRACTICE
+
PROJECTS
=
SUCCESS